Project Design Phase
Proposed Solution Template

Date	11 July 2026
Team ID	SWTID-2026-9508
Project Name	Credit Card Approval Prediction System
Maximum Marks	2 Marks

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Manual credit card approval is slow and inconsistent across bank officers, leading to delayed decisions, uneven risk assessment, and a poor experience for applicants.
2.	Idea / Solution Description	A Machine Learning-based web application that takes an applicant's demographic and financial details as input and instantly predicts whether the credit card application should be Approved or Rejected, using a model trained on historical approval data and served through a Flask web interface.
3.	Novelty / Uniqueness	Instead of relying on a single algorithm or fixed manual rules, multiple ML models (Logistic Regression, Decision Tree, Random Forest, XGBoost) are trained and compared on standard evaluation metrics, and the best-performing model is deployed - making the decision process both data-driven and justified by evidence.
4.	Social Impact / Customer Satisfaction	Applicants receive a fast, consistent decision instead of waiting days with no visibility. Bank officers get a reliable decision-support tool that reduces manual workload and removes officer-to-officer inconsistency, improving fairness for applicants.
5.	Business Model (Revenue Model)	For a bank or financial institution, the system reduces operational cost per application (less manual review time) and reduces risk-related losses from inconsistent manual decisions. It can be offered as an internal decision-support tool, or licensed as a SaaS credit-scoring module to smaller financial institutions.
6.	Scalability of the Solution	The prediction pipeline (preprocessing + model) is modular and can be retrained on larger or updated datasets as more applicant data becomes available. The Flask application can be containerized and deployed on cloud infrastructure (e.g., IBM Cloud) to handle a growing number of concurrent applications.